Guest Editorial: Special Issue on Graph Learning

GRAPH learning represents a dynamic and rapidly advancing field at the intersection of graph theory (also known as network science), machine learning, and artificial intelligence. Its unique ability to model complex relationships and interactions in a scalable and efficient manner makes it a cornerstone for scientific and technological advancements across diverse domains. Graph learning techniques have yielded groundbreaking progress in tackling real-world challenges, from anomaly detection and recommender systems to smart surveillance, traffic forecasting, disease control, and drug discovery. Despite its rapid emergence and successes, the field faces challenges in areas, such as fundamental theory and models, algorithms and methods, supporting tools and platforms, and real-world deployment and engineering.

This Special Issue on graph learning of IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS aims to capture the cutting-edge research in this fascinating domain. With over 350 submissions and a rigorous review process, we present 21 impactful works highlighting key topics that are pushing the boundaries of graph learning. We hope that the contributions within will inspire further innovation and collaboration, propelling the field towards new and exciting frontiers.

A. Graph Learning Foundations and Methodologies

In [A1], Li et al. address the unique challenges posed by heterophilous graphs. Traditional graph neural network (GNN) models often struggle with these graphs due to their inherent differences from homophilous graphs, necessitating more complex aggregation methods beyond the typical one-hop neighborhood. This paper presents an innovative approach through the development of Haar-type graph framelets, which are designed with properties such as permutation equivariance, efficiency, and sparsity to enhance deep learning tasks on graphs. The authors introduce a graph framelet neural network model, namely PEGFAN, which utilizes these graph framelets to achieve superior performance.

In [A2], Zhang and Kabuka propose MARML, a novel method for learning network representations in complex, layered data (often seen in biological systems). Current methods struggle to capture interactions between different types of entities within these multilayer networks. MARML addresses this by incorporating recurring motif patterns, alongside standard node attributes and network structure information. This allows the MARML to capture higher order connections across different hierarchies. The authors demonstrate the effectiveness of MARML through link prediction and differentiation tasks, ultimately improving our understanding of complex biological systems.

GNNs are powerful tools, but they often require a lot of labeled data for training, which can be expensive and time-consuming to obtain. In [A3], Ding et al. tackle this challenge in the context of semi-supervised learning, where only a small amount of labeled data is available. Existing methods struggle to capture complex relationships in graphs with limited labels. The authors propose a novel framework, augmented graph self-training (AGST), designed to enhance robustness in low-data contexts by incorporating structural and semantic augmentation modules. Their evaluations on semi-supervised node classification tasks highlight AGST's superior performance in scenarios with limited labeled data, showcasing its potential for sustainable graph learning.

Classifying rare data points accurately is a major hurdle in machine learning, as models often favor the majority class. In [A4], Ganaie et al. tackle this challenge by proposing GE-IFRVFL-CIL, a novel model that integrates graph embedding to preserve dataset topology and intuitionistic fuzzy theory to handle data uncertainty. By incorporating a novel weighting mechanism, GE-IFRVFL-CIL effectively addresses class imbalance, demonstrating superior performance on the Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset.

B. Graph Contrastive Learning

In [A5], Niu et al. shed light on limitations of current methods for hard negative mining in graph contrastive learning (GCL). While hard negative mining has proven effective in enhancing contrastive learning on various data types, its application to graphs has been hindered by issues such as oversmooth representations and non-independent and identically distributed (non-IID) data characteristics. This paper proposes an innovative method leveraging collective affinity information to identify hard negatives effectively in GCL. By integrating uncertainty measures into the loss functions, the approach enhances the discriminative power of learned graph representations, yielding substantial improvements in graph and node classification tasks across diverse datasets.

Knowledge graphs (KGs) are never complete. In [A6], Zeng et al. address critical challenges in enhancing the completeness of KGs through alignment techniques. Previous methods often focus solely on entity matching, neglecting relations and relying heavily on labeled data, which is scarce in practice. This paper introduces a versatile framework combining relation-enhanced active instance selection and cross-view contrastive learning to simultaneously align entities and relations under limited supervision. By selecting informative instances guided by relations and leveraging

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cross-view representations, the proposed framework significantly improves alignment performance across various KG pairs, showcasing its effectiveness in augmenting KG completeness efficiently.

In [A7], Wu et al. introduce MotifCC, a novel GCL framework for community detection in complex networks. Traditional community detection methods often focus on lower order structures, overlooking higher order connectivity patterns crucial to understanding network dynamics. MotifCC addresses this gap by leveraging motifs to construct a higher order network and employing contrastive learning to integrate diverse node, edge, and structural information effectively. By enhancing the fusion of higher and lower order features, MotifCC achieves superior community detection performance, demonstrated through extensive experiments on real-world datasets.

C. Graph Neural Networks

In [A8], Liu et al. address crucial challenges in GNNs when the underlying graph structure is unknown. Recent methods jointly learn the structure and GNN parameters but often assume a constant curvature for the embedding space (Euclidean or hyperbolic), leading to obfuscatory nodes that are poorly embedded and close to multiple categories. This paper introduces JSGL, a novel approach that first identifies and refines Euclidean obfuscatory nodes before refining their graph topology in hyperbolic space. The experimental results demonstrate JSGL's superiority over baseline methods, showcasing its effectiveness in improving GNN performance by addressing complex embedding space dynamics.

Accurately reconstructing time-varying graph signals is challenging. Traditional methods relying on smoothness assumptions and convex optimizations often face limitations in accurately capturing complex spatio-temporal dependencies. To address this issue, Castro-Correa et al. [A9] propose a novel approach using Gegenbauer-based graph convolutional operators. The proposed Gegenbauer-based time GNN (GegenGNN) enhances reconstruction accuracy through an encoder–decoder architecture and a dedicated loss function incorporating both fidelity to ground truth and signal smoothness properties. Experiments underscore GegenGNN's efficacy in recovering time-varying graph signals.

In [A10], Zhang et al. introduce a novel approach for solving complex optimization problems such as constraint satisfaction problems and combinatorial optimization problems. Traditional methods rely on branching heuristics, which can be problem-specific but complex, or general but potentially suboptimal. This work bridges the gap by introducing a solver framework that utilizes GNNs. The framework achieves competitive results on two NP-hard problems: the (minimum) dominating-clique problem and the edge-clique-cover problem.

D. Learning on Temporal, Spatial, and Complex Graphs

Temporal graphs, where connections change over time, are becoming a popular area of study. While existing methods achieve good results, they often lack transparency—we cannot understand how they adapt to new information. This is a major hurdle in scientific fields where interpretability is crucial. To address this issue, Peng et al. [A11] propose PiECL, a novel approach that combines physics-informed algorithms with continual learning principles to enhance explainability in temporal graphs. By quantifying data disturbance over time, PiECL provides transparent insights into the learning process, crucial for applications in chemistry, biomedicine, and beyond. The experimental results showcase the potential of PiECL to advance explainable temporal graph learning across diverse scientific domains.

Accurate forecasting is crucial for various applications using geo-distributed sensor networks, such as traffic prediction or pollution monitoring. Traditional methods struggle with the multivariate nature of data and spatio-temporal autocorrelation, often leading to suboptimal accuracy. In [A12], Altieri et al. proposes GAP-LSTM, a novel forecasting method that integrates graph convolution, attention-based long short-term memory (LSTM), 2-D convolution, and latent memory states. By synergistically leveraging these techniques, GAP-LSTM effectively captures complex spatio-temporal relationships across multiple nodes, thus improving forecasting accuracy in domains such as traffic, energy, and pollution.

While significant progress has been made in deep graph networks (DGNs), there is a pressing need to make DGNs suitable for predictive tasks on dynamic, real-world systems of interconnected entities. Gravina and Bacciu [A13] provide a comprehensive survey of recent advancements in learning temporal and spatial information, offering an in-depth overview of the current state of the art in dynamic graph learning. In addition, they conduct a rigorous performance comparison of popular approaches on node- and edge-level tasks, establishing a robust baseline for evaluating new architectures and methods.

E. Graph Learning for Anomaly Detection and Classification

Identifying anomalies in complex, interconnected data, such as multivariate time series, is crucial in various fields. Traditional methods often fall short in capturing nonlinear relationships and pairwise correlations among variables. In [A14], Zheng et al. propose a novel method, CST-GL, which leverages a correlation learning module to explicitly capture these correlations. By developing a spatio-temporal GNN, CST-GL encodes complex spatial dependencies and long-range temporal patterns using graph convolution networks and dilated convolutions. Extensive experiments validate CST-GL's effectiveness, showcasing its potential for early and accurate anomaly detection.

With the rise of Internet-of-Things (IoT) devices, efficiently detecting anomalies in their massive data streams becomes crucial, especially for resource-constrained devices. Zhou et al. [A15] present RG-GLD, a novel graph learning model that combines GNN with knowledge distillation to enable lightweight anomaly detection across IoT communication networks. The innovative graph network reconstruction strategy treats data communications as nodes in a directed graph, facilitating efficient and secure graph representation learning. Utilizing graph attention networks and multilayer perceptron techniques, RG-GLD effectively fuses structural and traffic features for enhanced anomaly detection in resource-limited IoT environments.

In [A16], Sun et al. tackle the challenge of classifying multilabel time series data, where data points have multiple relevant labels and the distribution of these labels is uneven. Existing methods struggle to capture the complex relationships between labels and address imbalanced data effectively. This research proposes a novel framework, DGAAE-MT, that leverages a dynamic graph attention autoencoder to model label relevance accurately. By employing a dual-sampling strategy and cooperative training, the framework enhances classification accuracy across all label frequencies.

F. Graph Learning Applications in Specific Domains

In [A17], Tang et al. deal with the complexities of analyzing electronic health records (EHRs) for precise healthcare decisions. Traditional federated graph learning methods face challenges due to the non-IID nature of EHRs, leading to data imbalance and reduced decision-making effectiveness. The proposed graph learning framework (called PEARL) addresses these issues by employing disease diagnostic code attention and admission record attention to extract patient embeddings. Integrating self-supervised learning within a federated framework, PEARL enhances disease prediction accuracy. In addition, a differential privacy scheme ensures data privacy. Through experiments on real-world datasets, PEARL demonstrates superior performance compared to existing methods.

In the field of drug discovery and material science, creating new molecules is crucial. Traditional generative models have focused on either 2-D bonding graphs or 3-D geometries, but not both simultaneously, limiting their effectiveness. In [A18], Huang et al. address this gap by proposing JODO, a novel diffusion model that considers both 2-D and 3-D aspects to generate complete molecules. Utilizing a diffusion graph transformer, JODO captures the relationships between molecular graphs and geometries, demonstrating superior performance in inverse molecule design and molecular graph generation on the QM9 and GEOM-Drugs datasets.

Inferring answers from multimodal contexts in textbook question answering (TQA) is not always easy. Traditional methods focus on intramodal semantics, neglecting the crucial intermodal relationships between text and diagrams. Zhang et al. [A19] address this challenge by introducing IMR-HGN, an intermodal relation-aware heterogeneous graph network that aggregates different modalities while learning features rather than representing them independently. Experiments demonstrate that IMR-HGN outperforms existing methods, paving the way for more informative TQA systems.

Accurately counting objects in images and videos is a growing area of research, but background noise can hinder performance. Guo et al. [A20] propose a novel group and graph attention network (GGANet) to tackle this challenge. GGANet leverages an encoder–decoder architecture, integrating a group channel attention (GCA) module and a learnable graph attention (LGA) module. The GCA module assigns attention factors to subfeatures, while the LGA module treats the feature map as a graph, representing diverse features as vertices and their interactions as edges. This dual-module approach effectively reduces background noise and improves counting accuracy.

Recognizing actions from videos using skeletal graphs is an active area of research, but existing methods require a lot of labeled data and struggle with missing information such as absent joints. Huang et al. [A21] present GRA, a novel approach that addresses these limitations. GRA utilizes self-training to generate high-quality labels for unlabeled data, reducing the dependency on large labeled datasets. In addition, it employs a representation alignment technique to compensate for missing data points in the skeletal graphs. Evaluations show that GRA significantly improves action recognition performance, especially in scenarios with limited labeled data and incomplete skeletal information.

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APPENDIX: RELATED ARTICLES

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